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Flowering, seed dispersal, seed predation and seedling recruitment in two pyrogenic flowering resprouters

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Abstract. A few resprouting plants in fire-prone environments have no local seed bank (soil or canopy) when a fire occurs. These species rely on post-fire flowering and the production of non-dormant seeds to exploit favourable post-fire establishment and growth conditions. For two such pyrogenic flowering species (*Doryanthes excelsa* Correa and *Telopea speciosissima* (Smith) R.Br.), we examined the timing of seed release, patterns of fruit production, seed dispersal, seed predation and seedling establishment following a fire in the Sydney region of south-eastern Australia. Both species took some 19 months after the fire to flower and the first seeds were released 2 years after the fire. *D. excelsa* flowered and fruited only once after the fire. For *T. speciosissima*, plants also flowered at least once more in the subsequent 5 years, but produced seed in only the first three post-fire flowering years. Fruit production differed between species, with fruiting individuals of *D. excelsa* producing fewer infructescences, similar numbers of follicles, but many more seeds per follicle than fruiting individuals of *T. speciosissima*. Ultimately, *D. excelsa* produced approximately six times as many seeds per m^2 and four times as many seeds per adult in one flowering season than *T. speciosissima* did after four flowering (three successful fruiting) seasons.

Seeds were passively dispersed from fruits borne 3–4 m (*D. excelsa*) or 1–2 m (*T. speciosissima*) above the ground. Most seeds were found within 5 m (*D. excelsa*) or 3 m (*T. speciosissima*) of parent plants. The primary seed shadow of both species was a poor predictor of the distribution of seedlings, with more seedlings occurring further from the adults than expected from the distribution of seeds. Seed loss to predators was high in both species in exclusion experiments where mammals had access to clumps of seeds (77–88%). It was variable and generally lower (8–65%) in experiments where seeds were not locally clumped. However, for *T. speciosissima*, at one site, some 65% of seeds were lost to mammals and invertebrates in these latter experiments. At this site, these losses appeared to influence subsequent recruitment levels, as very low seedling densities were observed.

For both species, germination of seedlings first occurred some 2.5–3 years after the passage of the fire. The percentage of seeds produced to seedlings successfully established was low in *D. excelsa* (2–3%) and more variable across sites and years in *T. speciosissima* (0–18%). Resultant post-fire seedling densities of *D. excelsa* (two sites) and *T. speciosissima* at one site were similar, but they were much lower at the *T. speciosissima* site that had high levels of seed predation. Both *D. excelsa* and *T. speciosissima* are amongst the slowest woody resprouting species to recruit seedlings after fire in south-eastern Australia and lag years behind species with soil or canopy seed banks.

Introduction

In fire-prone habitats, recruitment in plant species is frequently linked to the passage of a fire (Tyler 1995). This results from at least two main mechanisms. First, in species with soil seed banks, seed dormancy may be broken by the passage of fires (e.g. via heat, smoke, charred wood: Keeley 1991; Brown 1993; or interactions of these factors: Keith 1997; Morris 2000). Examples include many legumes (Jeffery *et al.* 1988; Auld and O'Connell 1991) and non-leguminous species (Brown 1993; Dixon *et al.* 1995). Second, in species with canopy seed banks, fires result in the mass opening of fruits and the release of seeds (Bond *et al.* 1984; Lamont *et al.* 1991).

However, a number of species have no available seed storage in either the soil or canopy at the time of the passage of a fire (Keith 1996). A few may rely on dispersal from unburnt areas, but most have mechanisms that allow established plants to survive fire via buried lignotubers, rhizomes or buds (protected meristems). These species usually flower soon after fire and then release seeds (Gill 1981; Frost 1984; Johnson *et al.* 1994; Bond and van Wilgen 1996; Carrington 1999). For species dependant on post-fire flowering for recruitment (pyrogenic flowering resprouters),

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the timing and magnitude of this flowering will influence opportunities for post-fire seedling establishment. The longer the interval between the passage of a fire and seedling recruitment, the more of the available resources such as light, nutrients and space, are likely to have been captured by other resprouting species or by those species with soil or canopy seed banks that can recruit in the immediate post-fire period.

Many individuals of pyrogenic flowering resprouting species in south-eastern Australia appear to have a single post-fire flowering (e.g. *Angophora hispida*, Auld 1986; *Blandfordia nobilis*, Johnson *et al.* 1994; *Lomatia silaifolia*, Denham and Whelan 2000; *Xanthorrhoea australis*, Gill and Ingwersen 1976, but see; *X. fulva*, Taylor *et al.* 1998). In such cases, this represents the only opportunity for recruitment between fires. For other species, flowering may extend over several years following a fire (e.g. *T. speciosissima*, Pyke 1983; Bradstock 1995; Goldingay 2000; *Xanthorrhoea australis*, Curtis 1998). However, the timing of recruitment in pyrogenic flowering resprouting species has been little studied and the degree to which each post-fire flowering event contributes to recruitment is unexplored.

In the post-fire environment, the impact of seed predation may be reduced due to lower predator abundance and also via predator satiation, in the case of serotinous seed release (Whelan 1995; Bond and van Wilgen 1996; Myerscough *et al.* 2000). However, the seeds of pyrogenic flowering resprouters are likely to be released after most seeds from serotinous species have germinated and, consequently, would not benefit from any predator satiation effects from serotinous seed release. In addition, populations of predators may have recovered by this time. All pyrogenic flowering resprouters are thought to release seeds in a non-dormant state (Bell *et al.* 1993; Bond and van Wilgen 1996) and hence form transient soil seed banks (*sensu* Parker and Kelly 1989).

In this study, we assessed post-fire reproduction and the patterns of seedling recruitment after a wildfire in 1994, in two resprouting species reliant on post-fire flowering for seed production (*Doryanthes excelsa* and *Telopea speciosissima*), from the Sydney area of south-eastern Australia. We examined (i) the timing of flowering and magnitude of seed production over 5 years after the fire; (ii) the pattern of seed and seedling dispersal; (iii) the level of post-dispersal seed predation; and (iv) the proportion of seeds released that resulted in successful establishment of new plants.

Methods

Study species

Doryanthes excelsa Correa (Gymea lily—Doryanthaceae) is a large perennial rosette plant with a bulbous rhizome and long broad linear leaves up to 3 m long. It is apparently capable of asexual reproduction by branching of the rhizome (Nash 1996). It grows in dry sclerophyll forests and woodlands on soils derived from sandstone containing some clay and is restricted to the Sydney Basin in south-eastern Australia, with three outlying populations in northern New South Wales (Harden 1990–1993). The species survives fire by having apical buds protected in underground leaf bases. It flowers in a large, red inflorescence held up to 5 m above the foliage of the plant (Harden 1990–1993). Fire appears to stimulate flowering, although sporadic flowering does occur in the absence of fire (Nash 1996). The mechanism determining flowering is unknown (Beeh 1996; Nash 1996).

Telopea speciosissima (Smith) R.Br. (waratah—Proteaceae) is a tall shrub, generally 2–4 m, which is also known for its striking red blooms that emerge above the resprouting shrubby vegetation in the post-fire environment. It is common in dry sclerophyll forest and woodlands on the central coast and central and southern tablelands of New South Wales (Harden 1990–1993). The species survives fire by having dormant buds protected in an underground lignotuber (Bradstock 1995). Aerial stems are completely regrown after each fire.

Sites

Three sample sites within a 10-km radius were chosen in Royal National Park, about 35 km south of Sydney (34°03′S, 151°03′E). All sites occurred on soils derived from the Triassic Hawkesbury Sandstone Formation, although Site 1 (with an ironstone influence) and Site 2 (with an alluvial influence) were on soils slightly more fertile than Site 3. At Site 1, both study species occurred in ironstone woodland vegetation (D. Keith and M. Tozer, unpubl. data), with an overstorey of *Eucalyptus sieberi*, *Corymbia gummifera*, *Banksia serrata*, *Xylomelum pyriforme* and *Ceratopetalum gummiferum*. The shrubby understorey included *Grevillea diffusa*, *Banksia spinulosa*, *Lissanthe strigosa*, *Prostanthera* sp., *Isopogon anemonifolius*, *Acacia myrtifolia*, *Xanthorrhoea media*, *Lomandra* spp. and *Dianella* sp. The vegetation at Site 2 (*D. excelsa* only) is Sandstone Gully Forest (D. Keith and M. Tozer, unpubl. data), with an overstorey of *Angophora costata*, *Banksia serrata*, *Ceratopetalum gummiferum*, *Eucalyptus piperita* and *E. sieberi* and a shrubby understorey that included *Boronia pinnata*, *Grevillea mucronulata*, *Acacia terminalis*, *Lomandra longifolia* and *Caustis flexuosa*. At Site 3 (*T. speciosissima* only) the vegetation is Sandstone Heath (D. Keith and M. Tozer, unpubl. data), with a short overstorey of *Corymbia gummifera*, *Eucalyptus obstans*, *E. sclerophylla* and *Banksia serrata* and a shrubby understorey that included *Lambertia formosa*, *Lissanthe strigosa* and sedges including *Xanthorrhoea media*, *Restio dimorpha* and *Cyathochaeta diandra*. Plant nomenclature follows Harden (1990–1993).

All sites were burnt in January 1994. Before 1994, there was variation in fire history across sites. The eastern part of Site 1 (sampled for *D. excelsa*) and Site 3 were burnt in 1988. The western part of Site 1, separated by a road from the eastern part (sampled for both study species), and Site 2 were burnt in the early 1980s (NPWS fire records; D. Keith, pers. comm.).

Timing of flowering and seed release

The three study sites were monitored to observe flowering after the January 1994 fire for both species up until December 2000. As both *D. excelsa* and *T. speciosissima* flower in July–October, this involved seven potential flowering seasons (1994–2000). General observations were also made throughout Royal National Park to determine whether the pattern of the study sites was representative of the general response of the study species to fire in this area. The timing of fruit development and seed release was recorded for the first four flowering seasons after the 1994 fire.

Estimates of seed production and seedling establishment

Six 30 × 30-m plots were established as follows: *D. excelsa*, two plots at each of Sites 1 and 2; *T. speciosissima*, one plot at each of Sites 1 and 3. For both study species, within each plot at each site, the numbers of infructescences and follicles on all plants were scored each year (1994–2000). To estimate the number of seeds per follicle, intact

follicles were collected in autumn 1996, prior to dehiscence, for each species from each site. For *T. speciosissima*, samples of 23 and 25 follicles were collected from Sites 1 and 3, respectively, while for *D. excelsa*, samples of 15 and 29 follicles were collected from Sites 1 and 2, respectively. Seeds within these follicles were scored after dehiscence in the laboratory. For *T. speciosissima*, only those seeds judged as 'filled' (i.e. likely to be viable) were included in this assessment (see also Bradstock 1995; Denham and Whelan 2000). For *D. excelsa*, no assessment was made, although Nash (1996) found that the average viability of seeds collected from a number of locations (not including Royal National Park) was 86% (±6.2 s.e.). The individuals used to estimate numbers of seeds per follicle were sampled from outside the plots to avoid impacts on primary seed dispersal and seedling recruitment. The density of seeds per plot was estimated by multiplying the number of intact follicles by the estimate of the mean number of seeds per follicle and dividing this value by the area of each plot (900 m^2).

To estimate the magnitude of post-fire seedling recruitment for both study species, five randomly allocated transects each 1 × 30 m were searched for seedlings in spring 1998, within each of the 30 × 30-m plots. For *T. speciosissima*, we assessed the probable fruiting year from which seedlings were established, by comparisons with other seedlings of known age within the sites.

Primary seed dispersal

The seeds of *D. excelsa* are disc-like, about 15 mm in diameter, 2 mm thick, have a dry spongy texture and weigh approximately 43 mg (Westoby *et al.* 1990; Nash 1996; A. J. Denham, pers. obs.). The seeds of *T. speciosissima* are winged and weigh approximately 72 mg (Westoby *et al.* 1990). Westoby *et al.* (1990) considered *T. speciosissima* to be adapted for wind dispersal, but could only suggest that the spongy texture of *D. excelsa* seed could serve for wind or water dispersal. Initial seed fall from dehisced follicles in both species is passive and movement in the air appears likely to be the main means of dispersal away from the infructescence.

The spatial pattern of initial seed fall was estimated by counting seeds found on the ground in four orthogonal transects along compass bearings from each of five isolated plants at Site 1 for *D. excelsa* and Sites 1 and 3 for *T. speciosissima*. Because of time constraints, this was not done at Site 2 for *D. excelsa*. Searches were done in June–August of 1996, when most seeds had fallen from the follicles. Some further re-distribution of seeds along the soil surface by wind and rain after this time is likely (Lamont *et al.* 1993; Hammill *et al.* 1998). The transects for *D. excelsa* started approximately 0.25 m from the centre of each clump, because of the space occupied by the leaf bases of the plants. Counts of numbers of seeds were made every 0.1 m along each transect. Transects were 0.5 m wide and limited to 7.25 m long as there were no seeds seen in initial searches beyond 7.15 m. For *T. speciosissima*, the area occupied by the stems was included in the transects. In preliminary searches around *T. speciosissima* adults, no seeds were found more than 3 m from adults at Site 1 and 2 m from adults at Site 3, so the transect lengths were reduced to 3.5 and 2.5 m, respectively.

Seed predation

The fates of recently released seeds on the ground were examined during the initial seed release after the 1994 fire (i.e. in 1996), in two trials. First, an exclusion experiment was used to investigate the agents responsible for and the degree of post-dispersal predation at each site. It was assumed that no secondary dispersal of seeds by invertebrates such as ants was likely, since neither species have lipid bodies attached to their seeds. Westoby *et al.* (1990) suggested that neither species is myrmecochorous and field observations supported this. A block design was used with two treatments and 10 seeds per treatment per block. This experiment was set up for *D. excelsa* at Sites 1 and 2 and for *T. speciosissima* at Sites 1 and 3 as follows:

(a) caged: cages (15 × 15 cm × 5 cm high with 1-cm^2 mesh size) were used to exclude vertebrate predators, while still allowing access to most invertebrates. To prevent seeds from washing away in heavy rain, a PVC ring sawn from pipe 10 cm in diameter and about 1 cm tall was placed on the ground. This ring was pushed into the soil leaving only the top 0.5 cm protruding. The cage was placed over the ring and fastened down with wire stakes.

(b) uncaged: where only the PVC ring was used. This allowed access to both invertebrates and vertebrates.

Ten blocks consisting of a caged and an uncaged treatment, were used for each species, at each site. The blocks were placed about 5 m apart along a transect through the site, with separate transects for each species at Site 1.

The number of seeds remaining in each treatment block was recorded every day for the first week and approximately every second day for the following 10 weeks. Additionally, any information concerning the condition of seeds (such as whether chewed or germinated) or the presence of testa remains was recorded at each visit.

Second, for both study species, a trial was established to examine the level of predation on seeds that were placed singly at random locations throughout the plots, rather than clumped as above (see Auld and Denham 1999, 2001). These trials were established for *D. excelsa* in two plots at Site 1 and in one plot at Site 2 and for *T. speciosissima* in one plot at each of Sites 1 and 3. A single seed was placed at each of 10 randomly located positions according to the 30 × 30-m grid established at each site, with the location marked by a steel stake 0.2 m away. Additional batches of 10 seeds were similarly placed in new random locations each week for the following 3 weeks. The fate of these seeds was followed for the next 10 weeks, with details recorded as for the exclusion experiment. Only 'filled' seeds were used in these experiments.

Seedling dispersal

The following two different estimates were made of the dispersal pattern of seedlings around adult plants:

(a) the ground was searched for seedlings up to a distance of 10 m from fruiting plants around five isolated plants of *D. excelsa* at Site 1. The distance of each seedling from the centre of the adult was measured. The adults included two of those used to sample primary seed dispersal. Sufficient seedlings were not available at the other three adults sampled for primary seed dispersal. For *T. speciosissima*, this technique was not used as adults continued to flower after the first year.

(b) for *D. excelsa* at Sites 1 and 2 and *T. speciosissima* at Sites 1 and 3, the distance of each seedling from the centre of the nearest parent was measured for all seedlings observed in the transects used for estimating seedling establishment (above). Because of the overall low number of seedlings observed for *T. speciosissima* at Site 3, the entire 30 × 30-m plot was surveyed. This estimate included overlapping seed shadows from adult plants, as typically occurs in the field.

Analyses

To test the hypothesis that fruit and seed production varied across sites and years within the study area, estimates of infructescence and follicle production per fruiting plant were compared in *D. excelsa* by a nested 2-factor ANOVA with plots nested within sites and in *T. speciosissima* across sites and years when fruit was produced, by a 2-factor ANOVA without replication. Estimates of seed production per m^2 were compared in *D. excelsa* across sites by a 1-factor ANOVA with plots as replicates and in *T. speciosissima* across sites and fruiting years by a 2-factor ANOVA without replication (Zar 1996).

To test the hypothesis that vertebrate seed predators had an impact on the number of seeds available for germination (the seed predation

exclusion experiment), the proportions of seeds remaining in each replicate treatment within the 10 blocks after 75 days were compared by a 3-factor ANOVA (factors were treatment, site and block, with blocks nested within sites) for each species separately. The hypothesis that the two species suffered different levels of predation at Site 1, where both *D. excelsa* and *T. speciosissima* were sampled, was tested by a 3-factor ANOVA with species and caging treatment (with blocks nested within species) as factors.

At lower seed densities than in the seed predation exclusion experiment (the single seeds trial) we tested the hypothesis that predation varied across sites. For *T. speciosissima*, the proportions of seeds remaining in each weekly cohort of 10 seeds at the end of the study (after 9 weeks) were compared across sites by a 1-factor ANOVA. Data for *D. excelsa* were similarly analysed, using both plots at Site 1 and the single plot at Site 2 as three levels in a 'location' factor. Again, it was possible to test the hypothesis that the two species suffered different levels of predation at Site 1 at this lower seed density. Here, only the western of the two *D. excelsa* plots was used, as it was adjacent to the *T. speciosissima* plot (1-factor ANOVA).

We tested the hypothesis that seedling establishment varied across sites in both species and across years for *T. speciosissima*. In *D. excelsa*, the number of seedlings per transect was compared across sites by a nested 2-factor ANOVA with plots nested in sites. For *T. speciosissima*, the number of seedlings per transect was compared across sites and fruiting year of origin by a 2-factor ANOVA.

For each ANOVA, homogeneity of variances was tested by Cochran's test. Where variances were significantly heterogeneous an arcsine transformation was applied to proportion data and a square-root transformation was applied to count data. If variances were still heterogeneous, the analysis was carried out on the raw data with a conservative approach to rejection of the null hypothesis (Underwood 1981).

To test the hypothesis that the distributions of seedlings around plants were the same as the distributions of seeds around plants, a number of Kolmogorov–Smirnov goodness-of-fit tests were done.

(i) For *D. excelsa* at Site 1, seed and seedling data from the isolated plants were placed into 0.1-m classes and frequency counts of seeds or seedlings were obtained from each distance class for all five individuals. The proportions of seeds or seedlings per distance class were then determined. Seedling data for distances less than 0.25 m from the adult were omitted, since data for seed distribution were not obtained at this distance, due to the occupation of most of the available space by the parent plants. Similarly, seedling distances greater than 7.15 m were omitted since this was distance of the furthest observed seed. A Kolmogorov–Smirnov test was used to determine the goodness-of-fit of the seedling distribution (observed) with the seed distribution (predicted) (Zar 1996).

(ii) For *D. excelsa* at Site 1 and *T. speciosissima* at Sites 1 and 3, estimations were made of the goodness-of-fit of the seedling distributions from the plot transect sampling (observed) with the seed distributions from the isolated individuals (expected). Restriction of seedling data to the greatest distance observed for seeds (7.15 m) was as above for *D. excelsa*, and to 3.75 m for *T. speciosissima*. There was no restriction on the least distance for *T. speciosissima*. This test enabled the determination of whether the distribution of seeds around isolated adults adequately explained the distribution of seedlings around adults that were contagiously distributed (a more typical situation) and thus had overlapping seed shadows.

(iii) For *D. excelsa* at Site 1, the seedling distribution from the five isolated plants ('expected') was compared with that obtained from the plot transect sampling ('observed'). This test assessed how well the distribution of seedlings around isolated plants of *D. excelsa* represented the more typical situation of contagiously distributed individuals.

Results

Timing of flowering and seed release

For both *D. excelsa* and *T. speciosissima*, the first flowering after the January 1994 fires occurred in late winter (August) of 1995, some 19 months after the January 1994 fire. This pattern was found at all study sites and more generally across Royal National Park. Fruits matured in the following autumn (March–June 1996), with *D. excelsa* first releasing seeds in March 1996 and *T. speciosissima* in April 1996. Consequently, seeds were not released into the post-fire habitat until some 25–26 months after the passage of the January 1994 fire. Seed fall was quite prolonged with some seeds still in fruits in June (both species). *D. excelsa* plants in the study plots did not flower in any of the subsequent 5 years, while most *T. speciosissima* individuals flowered again the following year and a few in three of the subsequent 4 years (Table 1), with a similar timing of flowering and fruiting to that observed in 1995–1996.

Fruit and seed production

In *D. excelsa*, the number, and hence density, of flowering plants varied considerably between plots within sites, ranging from six at Site 1, plot 1 (0.2 m^{-2}) to 39 at Site 2, plot 2 (1.3 m^{-2}) (Table 1). There were no significant differences between sites in the mean numbers of infructescences ($F_{1,2} = 0.86$, $P = 0.45$) or follicles ($F_{1,2} = 9.94$, $P = 0.09$) produced per fruiting plant, although there were almost twice as many follicles produced at Site 1, plot 1, than at Site 2, plot 2 (Table 1). For *T. speciosissima*, the total number of individuals that flowered between 1995 and 2000 varied from 14 at Site 1 to 24 at Site 3 (Table 1). There was no significant difference in the mean number of infructescences ($F_{1,2} = 0.4$, $P = 0.59$) or follicles ($F_{1,2} = 0.003$, $P = 0.96$) per fruiting plant between sites across the 1995–1997 flowering seasons. *D. excelsa* had greater seed production per follicle than *T. speciosissima* (104, 105 *v.* 11 seeds per follicle, respectively, Table 2). There were no significant differences in the number of seeds per follicle between sites for *T. speciosissima* ($F_{1,46} = 0.43$, $P = 0.52$) or *D. excelsa* ($F_{1,42} = 0.08$, $P = 0.93$) (Table 2). The estimated density of seeds produced by *D. excelsa* varied greatly, although not significantly ($F_{1,2} = 4.76$, $P = 0.16$) across sites, with Site 1 producing 9.9 (±0.4 s.e.) seeds per m^2 and Site 2 producing 17.7 (±3.6 s.e.) (Table 3). The lack of significance in this comparison is likely to be a result of the considerable variation between plots at Site 2, with one plot producing approximately 1.5 times as many seeds as the other. This in turn was largely because of the much higher density of flowering and fruiting plants in plot 2 than in plot 1 of Site 2 (Table 1). There were no significant differences in seed density of *T. speciosissima* between sites ($F_{1,2} = 1.11$, $P = 0.40$) or among years ($F_{2,2} = 4.19$, $P = 0.19$). In the first post-fire fruiting year (1996), the estimated density of seeds

Table 1. Estimates of reproductive output in *Doryanthes excelsa* and *Telopea speciosissima*
Data for infructescences and follicles are means per fruiting adult (standard errors are given in parentheses). There was no flowering in the 1994 flowering season for either study species

Site	Plot	Season (year)	No. of flowering plants	No. of fruiting plants	Mean no. of infructescences per fruiting plant	Mean no. of follicles per fruiting plant
				D. excelsa[A]		
1	1	1995	6	6	1.5 (0.34)	15.2 (7.0)
1	2	1995	10	7	1.7 (0.36)	12.0 (2.9)
2	1	1995	19	6	1.7 (0.33)	9.2 (2.4)
2	2	1995	39	20	1.8 (0.18)	8.2 (1.5)
				T. speciosissima		
1	1	1995	12	10	2.0 (0.5)	8.6 (3.0)
1	1	1996	13	10	2.0 (0.4)	5.6 (2.8)
1	1	1997	6	6	1.8 (0.3)	2.3 (0.8)
1	1	1998	2	0	0	0
1	1	1999	0	0	0	0
1	1	2000	0	0	0	0
1	1	1995–2000	14	14	3.6 (0.8)	11.1 (4.7)
3	1	1995	19	15	1.7 (0.2)	5.9 (1.0)
3	1	1996	20	16	2.4 (0.6)	8.4 (3.3)
3	1	1997	7	7	1.1 (0.1)	2.6 (0.6)
3	1	1998	2	0	0	0
3	1	1999	1	0	0	0
3	1	2000	0	0	0	0
3	1	1995–2000	24	21	3.4 (0.6)	11.4 (2.9)

[A]*Doryanthes excelsa* only produced fruit in one year.

produced was 1.1 seeds per m^2 at both sites (Table 3). Seed production was higher in the second fruiting year at Site 2, but lower at Site 1. Despite the lack of significance of a year effect in the analysis, at both sites seed production declined dramatically by the third fruiting year (1997) and no seeds were produced in subsequent years (Table 1). Total post-fire seed production by *D. excelsa* was considerably greater than by *T. speciosissima* (e.g. 9.9 *v.* 2.0 seeds per m^2, 2350 *v.* 430 seeds per flowering plant for *D. excelsa* and *T. speciosissima*, respectively, at Site 1) (Table 3).

Seed dispersal

Seed dispersal approximated an exponential decay curve (see also Janzen 1970; Okubo and Levin 1989; Willson 1992). Most seeds fell within 5 m of adults for *D. excelsa* and within 3 m of adults for *T. speciosissima* (Fig. 1).

Table 2. Estimates of the number of seeds per follicle in *Doryanthes excelsa* and *Telopea speciosissima*
Data are means ± standard errors

Species/site	No. of follicles sampled	Seeds per follicle
D. excelsa/1	29	103.7 (10.2)
D. excelsa/2	15	105.3 (18.5)
T. speciosissima/1	25	11.4 (0.4)
T. speciosissima/3	23	11.0 (0.4)

Seed predation

Seed losses to predators were of two types, with consumption of whole seeds (occasionally all 10 seeds in a replicate) and partial to complete seed damage *in situ.* Consumption of whole seeds was most likely to have been by

Table 3. The estimated density of seeds released by *Doryanthes excelsa* and *Telopea speciosissima*, the density of seedlings in spring 1998 and the proportion of seeds (%) that had established as seedlings by spring 1998 (4.5 years post-fire)

Site	Year of seed release	Seeds per m^2 (s.e.)[A]	Seedlings per m^2 (s.e.)	Establishment (%)
		D. excelsa		
1	1996 (total post-fire)	9.9 (0.4)	0.31 (0.15)	3.1
2	1996 (total post-fire)	17.7 (3.6)	0.28 (0.08)	1.6
		T. speciosissima		
1	1996	1.1	0.12 (0.03)	19.3
1	1997	0.7	0.15 (0.07)	20.6
1	1998	0.2	0.02 (0.01)	11.2
1	Total post-fire	2.0	0.29 (0.09)	14.1
3	1996	1.1	0.02 (0.01)	1.9
3	1997	1.6	0.01 (0.01)	0.4
3	1998	0.2	0	0
3	Total post-fire	2.9	0.03 (0.01)	0.7

[A]No standard errors on *Telopea speciosissima* as there was no replication within sites.

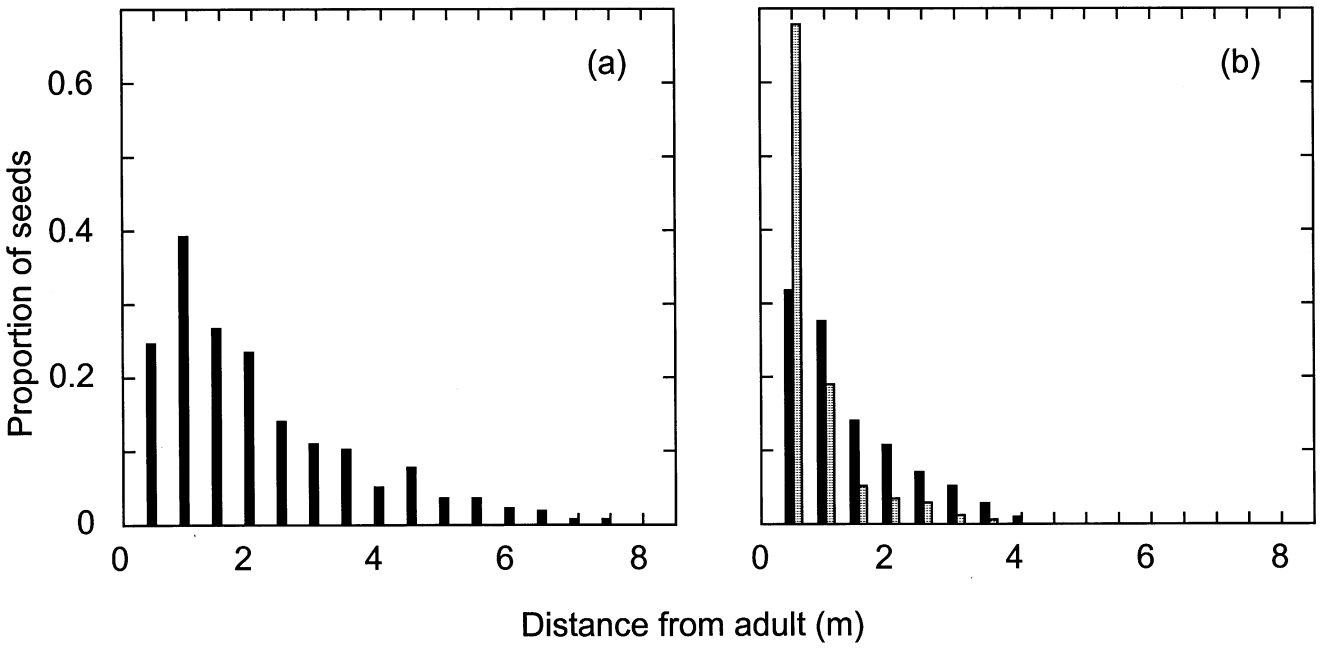


Fig. 1. Seed dispersal around adults for (*a*) Site 1, *Doryanthes excelsa*, and (*b*) Sites 1 and 3, *Telopea speciosissima*. Bars represent proportions of the total of all seeds found within four 0.5-m-wide transects around five isolated individuals at each site: Site 1 (filled bars), Site 3 (stippled bars). Total number of seeds found was 924 for *D. excelsa* and 214, 173 for *T. speciosissima* at Sites 1 and 3, respectively.

mammals such as swamp wallabies (*Wallabia bicolor*) and probably introduced Rusa deer (*Cervus timorensis*) that occur in the habitat. We suggest that *in situ* seed damage was most likely to have been caused by rodents such as *Rattus fuscipes* and invertebrates as indicated by the pattern of seed remains and bite marks on seeds (see also Auld and Denham

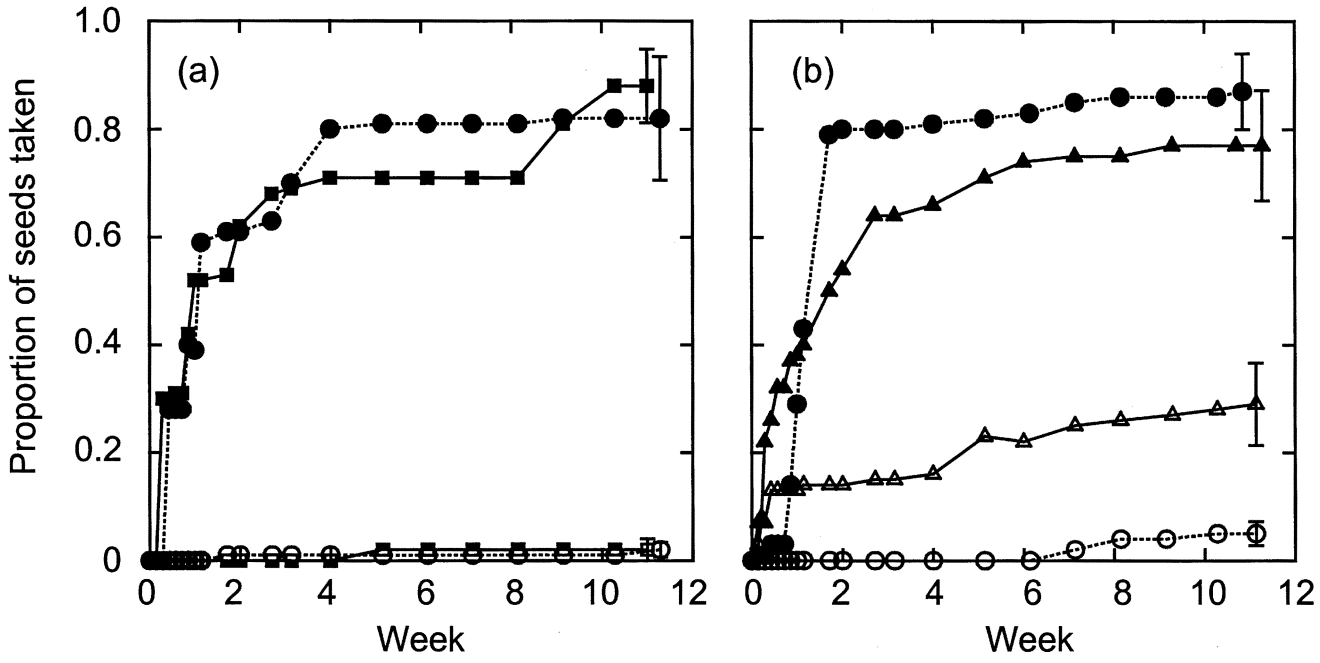


Fig. 2. Rate of seed predation in exclusion experiment: (*a*) *Doryanthes excelsa* and (*b*) *Telopea speciosissima*. Site 1 (circles), Site 2 (squares), Site 3 (triangles), caged treatment (open symbols), uncaged (filled symbols). Standard errors are based on Week 11 (week values for some points juggled for clarity).

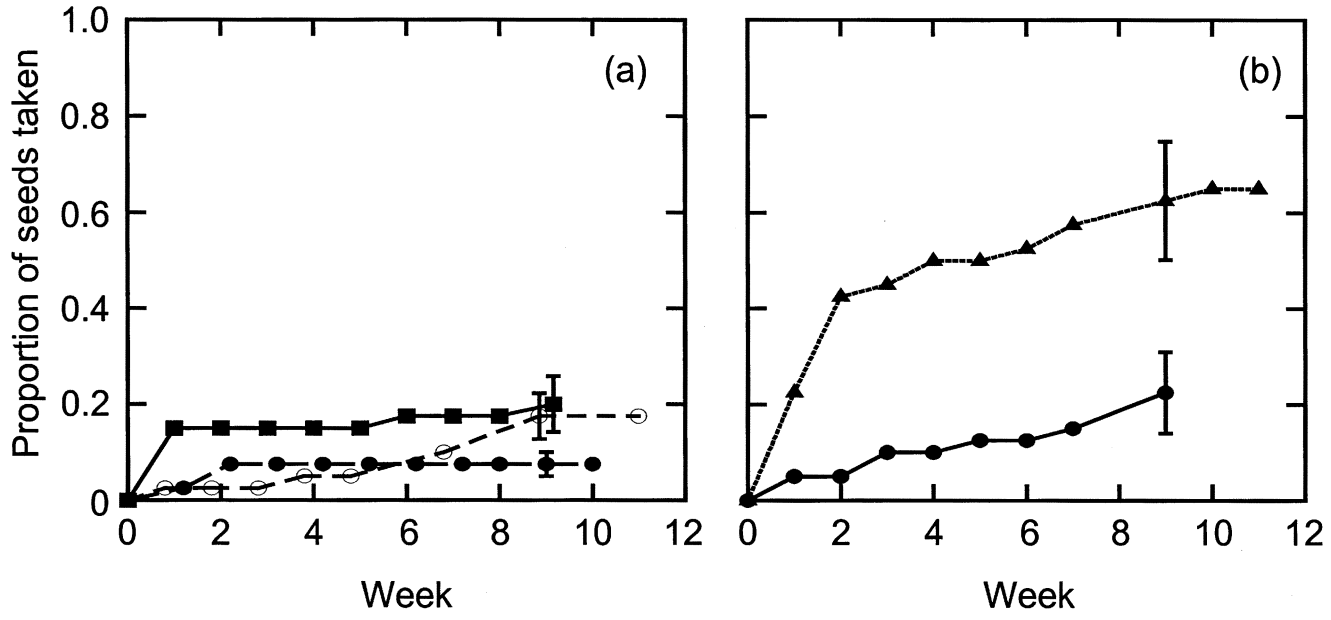


Fig. 3. Rate of predation of single seeds: (*a*) *Doryanthes excelsa* and (*b*) *Telopea speciosissima*. Site 1 (circles), Site 2 (squares), Site 3 (triangles), western (filled circles) and eastern plots (open circles) of *D. excelsa* at Site 1, respectively. Standard errors are based on Week 9.

1999). In the exclusion experiment, predation on uncaged seeds was high. For sites pooled, 85% (±6.5 s.e.) of seeds were taken after 11 weeks in *D. excelsa* and 82% (±6.1 s.e.) in *T. speciosissima* (Fig. 2). In contrast, few seeds were removed from or eaten in the caged treatment, with 2% (±1.2 s.e.) of seeds taken after 11 weeks in *D. excelsa* and 17% (±4.8 s.e.) in *T. speciosissima* (Fig. 2). Invertebrates, small enough to fit through the mesh, presumably ate these latter seeds. For *D. excelsa*, despite the persistence of heterogeneous variances (Cochran's test), we consider the treatment effect to be significant ($F_{1,1}$ = 765.4, P = 0.023), given the robust nature of ANOVA and the large difference in means (85% ± 6.5 s.e. cf. 2% ± 1.2 s.e.) (Underwood 1981) (Fig. 2*a*). For *D. excelsa*, no interaction or site effects were significant. For *T. speciosissima*, the main treatment effect was not significant ($F_{1,1}$ = 14.62, P = 0.16). However, there was a significant ($F_{1,18}$ = 8.45, P = 0.01) treatment × site interaction for *T. speciosissima*, resulting from higher mean seed losses from the caged treatment at Site 3 than at Site 1 (Fig. 2*b*). This suggests a greater impact of invertebrate predation at Site 3. However, pairwise comparisons indicated only that the caged treatments were significantly different from the uncaged treatments, with no significant differences between sites within treatments. Where both study species were compared at Site 1, there was a significant ($F_{1,1}$ = 6561, P = 0.01) treatment effect, with the pattern for both species being similar, i.e. caging significantly reduced seed-predation levels.

With the exception of Site 3 for *T. speciosissima* (Fig. 3*b*), seed losses to predators in the single-seed trial were much lower than in the uncaged treatment of the exclusion experiment, where seeds were clumped at a much higher density (Fig. 2 *v.* Fig. 3). For *D. excelsa*, while there was no significant ($F_{2,9}$ = 1.23, P = 0.34) site effect, the greatest

Table 4. Analysis of the spatial distributions of seeds and seedlings for *Doryanthes excelsa* and *Telopea speciosissima*

Comparison	Max. difference (D_{max})	Significant D_{max}	P-value
D. excelsa seeds *v.* seedlings at Site 1 (isolated plants only)	0.509	$D_{0.05,160}$ = 0.106	<0.001
D. excelsa seeds (isolated plants) *v.* seedlings (non-isolated plants—transect sampling in plots) at Site 1	0.431	$D_{0.05,86}$ = 0.144	<0.001
D. excelsa seedlings (isolated plants) *v.* seedlings (non-isolated plants—transect sampling in plots) at Site 1	0.301	$D_{0.05,90}$ = 0.141	<0.001
T. speciosissima seeds (isolated plants) *v.* seedlings at Site 1 (non-isolated plants—transect sampling in plots)	0.533	$D_{0.05,43}$ = 0.203	<0.001
T. speciosissima seeds (isolated plants) *v.* seedlings at Site 3 (non-isolated plants—transect sampling in plots)	0.515	$D_{0.05,72}$ = 0.158	<0.001

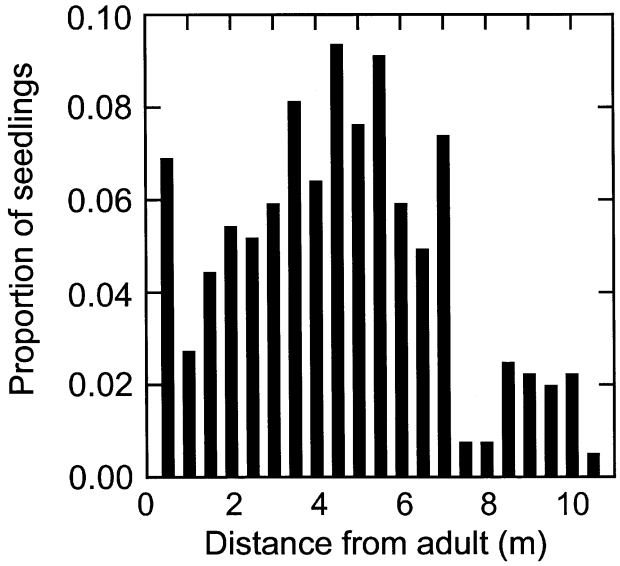


Fig. 4. Seedling dispersal around isolated adults at Site 1 for *Doryanthes excelsa*. Bars represent proportions of all seedlings found within 0.5-m-wide distance interval classes. Data pooled from five individuals.

losses were at Site 2 (mean of 20%), compared with mean losses per plot of 7.5 and 17.5% at Site 1 (Fig. 3*a*). For *T. speciosissima*, there was a significant ($F_{1,6}$ = 7.81, P = 0.03) site effect, with a greater loss of seeds from Site 3 (mean 65%) than from Site 1 (22.5%) (Fig. 3*b*). At Site 1, there was a significant difference in the level of predation between study species ($F_{1,6}$ = 22.09, P = 0.003), with a greater percentage of seeds removed for *T. speciosissima* (22.5%) than for *D. excelsa* (7.5%, western plot only) (Fig. 3). Several of the seeds of both species still present 9 weeks after placement had germinated, indicating that they had not escaped predation by virtue of being inviable. Some of these were established as seedlings by spring 1998.

Seedling establishment

The first seed germination event after the 1994 fire for both species was observed in June 1996, some 2.5 years after the fire. Seedlings were found at all sites. The density of seedlings that had established and persisted up to 1998 for *D. excelsa* varied from 0.28 (± 0.08) per m^2 at Site 2 to 0.31 (± 0.15) per m^2 at Site 1 (Table 3). There were no significant differences in seedling densities between sites ($F_{1,16}$ = 0.02, P = 0.88) or plots within sites ($F_{1,16}$ = 0.47, P = 0.63). For *T. speciosissima*, the density of seedlings established by 1998, from any one post-fire seed-crop, ranged from 0 per m^2 at Site 3 following 1998 fruiting to 0.15 per m^2 ± 0.07 at Site 1 following 1997 fruiting, although fruiting year did not significantly ($F_{2,2}$ = 1.44, P = 0.41) influence seedling establishment (Table 3). For all fruiting seasons combined, there were approximately 10 times as many seedlings

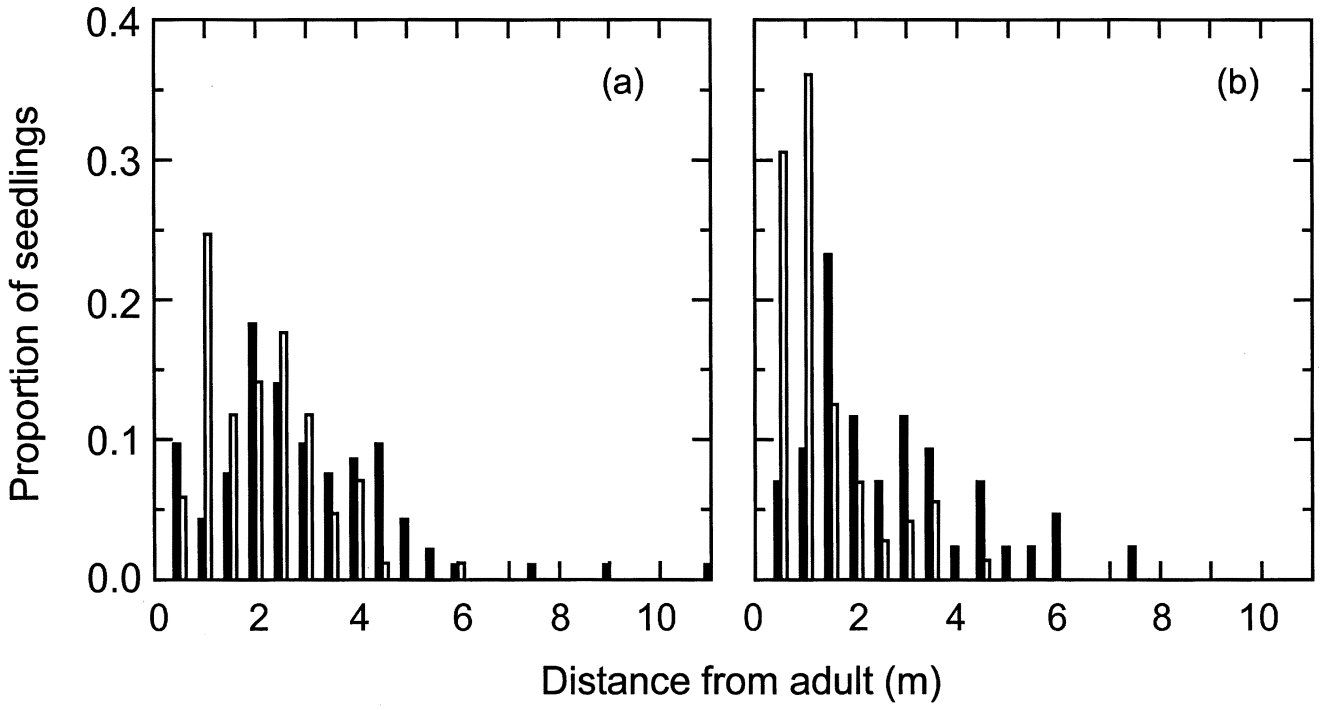


Fig. 5. Seedling dispersal around non-isolated adults at (*a*) Sites 1 and 2 for *Doryanthes excelsa* and (*b*) Sites 1 and 3 for *Telopea speciosissima*. Bars represent proportions of all seedlings pooled from five transects for each site, except Site 3 where they represent seedlings pooled from the entire 30 × 30-m plot. Site 1 (*a* and *b*, filled bars), Site 2 (*a*, unfilled bars) or Site 3 (*b*, unfilled bars).

Table 5. Summary of literature giving time to recruitment of seedlings after fire for species of different seed bank types from various fire-prone environments

Results from this study for *Doryanthes excelsa* and *Telopea speciosissima* (>2.5 years) show that these species are considerably slower at post-fire seedling recruitment

Seed bank type	Location of study species	Period from fire to seedling recruitment	Source
Soil seed bank	Australia	<1 year	Purdie (1977), Hughes and Westoby (1992), Meney *et al.* (1994), Auld and Tozer (1995), Benwell (1998)
	South Africa	<1 year	Bond *et al.* (1984), Kruger and Bigalke (1984)
	USA	<1 year	Keeley (1991)
Canopy seed bank	Australia	<1 year	Bradstock and Myerscough (1981), Bradstock and O'Connell (1988), Lamont and Barker (1988), Lamont *et al.* (1991), Benwell (1998)
	USA	<1 year	Keeley *et al.* (1999), Turner *et al.* (1999)
Post-fire flowering (ephemeral or no seed bank)	Australia	<2 years	Auld (1986), Benwell (1998), Curtis (1998), Taylor *et al.* (1998)
	USA	<2 years	Carrington (1999), Brewer and Platt (1994), Malanson and O'Leary (1982), Keeley and Keeley (1984)
	South Africa	<2 years	Kruger (1983)

recruited at Site 1 than at Site 3 in *T. speciosissima* ($F_{1,24} = 11.1$, $P = 0.003$, Table 3).

The percentage of the seeds released that established as seedlings varied between species, sites and years. For *D. excelsa*, both sites had low establishment percentages (1.6–3.1%) (Table 3). For *T. speciosissima* at Site 3, the percentage seedling establishment was low (0–1.9%), but comparable to *D. excelsa*, while at Site 1 percentage seedling establishment was higher (11.2–20.6%) (Table 3). The resultant seedling densities per m^2 were similar for both species, except for *T. speciosissima* at Site 3, where they were an order of magnitude lower (Table 3). For *T. speciosissima*, these seedling densities reflected an accumulation of cohorts of seedlings from 3 years of post-fire fruiting, compared with 1 year for *D. excelsa*. In terms of seedlings per flowering plant there was more variation, with results for *D. excelsa* ranging from 64.5 seedlings per flowering adult at Site 1, plot 1 to 6 at Site 2, plot 2. For *T. speciosissima* for the three fruiting seasons combined, the range was from 22.7 seedlings per flowering adult at Site 1 to 0.1 at Site 3.

Seedling distribution

The distribution of seedlings around adults differed significantly from that of seeds (Table 4). In all cases, a greater proportion of seedlings occurred further from the adult than suggested by the seed distributions. When the distribution of seeds was compared with that of seedlings around isolated individuals of *D. excelsa* at Site 1, there was a difference in the shape of the respective distributions, with the distribution of seedlings fairly uniform up to about 7 m from an adult (Fig. 4). In contrast to the distribution of seeds (Fig. 1*a*), there was not a large proportion of seedlings near adults. These data suggest that seedlings had a similar likelihood of occurring anywhere up to 7 m from their parent. Where there were overlapping seed shadows from parent plants in *D. excelsa*, the distribution of seedlings (Fig. 5*a*) was uneven, but more or less uniform in density up to 4 m from a parent plant, with a decline further away. For *T. speciosissima*, the overall shape of the distributions of seeds and seedlings was similar, despite the significant difference in the distributions. The greatest proportion of seedlings occurred further from the adult than for seeds and the tail extended further from the adult (Figs 1*b*, 5*b*). The maximum difference in distributions tended to occur within 2 m from adults, suggesting that there is some interaction between adults and seedling recruitment that leads to reduced recruitment less than 2 m from adults. This effect appeared to be most pronounced in *T. speciosissima* at Site 1. In both species, recruits were located well beyond the limit of seeds detected in the seed-distribution surveys, suggesting either occasional long-distance dispersal directly from the infructescence or later (i.e. after our survey) dispersal occurring at ground level (Harper 1977; Hammill *et al.* 1998).

Discussion

Both pyrogenic flowering, resprouting species took some 19 months to flower after the January 1994 fire. For *D. excelsa*, this represented the only post-fire flowering event, although sporadic flowering in the absence of fire does occur in the species (Nash 1996). A single post-fire flowering event has been observed in several other local taxa that are pyrogenic flowering resprouters (Auld 1986; Denham and Whelan 2000), as well as from elsewhere, although most are herbaceous or geophytes (Frost 1984; Johnson *et al.* 1994; Bond and van Wilgen 1996). In *D. excelsa*, following the senescence of the flowering spike, individuals undergo subterranean branching, which may result in asexual reproduction (Beeh 1996; Nash 1996). This additional resource expenditure may explain the single post-

fire flowering event in *D. excelsa*. In *T. speciosissima*, there was further flowering in some individuals over the next 4 years after the fire, although no seeds were produced following flowering in 1998 (4.5 years post-fire) or later in our study plots. In *T. speciosissima*, there was a trend for the highest seed production in the first two post-fire fruiting seasons, followed by a marked decline in the third. Bradstock (1995) and Pyke (1983) observed a similar trend; however, Pyke (1987) suggested that temporal patterns of flowering may depend on other, unknown factors. Goldingay (2000) presented data indicating that for plants in one population of *T. speciosissima*, flowering was prolific for up to 8 years after fire, before declining subsequently. The temporal pattern of post-fire flowering in *T. speciosissima* clearly varies with location. Despite flowering only once after the 1994 fire, *D. excelsa* was very successful in producing seed, releasing nearly five times as many seeds as *T. speciosissima* at Site 1 where they co-occurred.

Both species took some 2.5 years after the fire to establish seedlings. Other species either cohabiting with *D. excelsa* and *T. speciosissima*, or from other fire-prone habitats, established seedlings much sooner. In general, those species with a soil seed bank mostly establish seedlings within the first year after a fire, while those with a canopy seed bank mostly release seed in the first 6 months (Lamont and Barker 1988; Lamont *et al.* 1991) and establish seedlings in the first year (Table 5). Even other species that rely on post-fire flowering for seedling establishment appear to be 0.5–1.5 years quicker than *D. excelsa* and *T. speciosissima* in establishing seedlings post-fire (Table 5). Among other Australian species, Auld (1986) found that for the shrub *Angophora hispida*, establishment occurred within about 18 months, while seedlings of shrubby *Xanthorrhoea* spp. were observed 18 months to 2 years after fire (Benwell 1998; Curtis 1998; Taylor *et al.* 1998). This is also the case in other fire-prone areas of the world, including Californian sage scrub, Florida sand pine scrub and South African fynbos (Table 5). By the time seedlings of *D. excelsa* and *T. speciosissima* are becoming established, much of the space, light and nutrients made available by the fire may have been captured by other species (both resprouting plants and seedlings). This will be particularly the case for seedlings from the second and third fruit crops post-fire in *T. speciosissima*. In other vegetation types, resprouting species may be able to continue recruiting new seedlings several years after fire (e.g. *Garberia heterophylla* in Florida, Abrahamson and Abrahamson 1996). Curtis (1998) observed seedlings of the arborescent monocot, *X. australis*, at sites unburnt for about 100 years. However, such inter-fire recruitment has rarely been observed in the post-fire environment of *D. excelsa* and *T. speciosissima*. Thus, despite the tolerance of their adults to fire, the persistence of these species may in part be due to their large seed size, which ensures that seedlings have sufficient reserves to establish (Westoby *et al.* 1992; Lamont and Groom 1998), although the relative proportions of different nutrients in these seeds has not been determined (Lamont and Groom 1998; Myerscough *et al.* 2000).

Primary seed dispersal for both species produced a high concentration of seeds adjacent to the adult and an exponential decay of seed density with distance. This pattern is typical of a broad range of other plant species (Janzen 1970; Auld 1986; Willson 1992; Hammill *et al.* 1998). However, Hammill *et al.* (1998) found that seeds from species with canopy seed banks were dispersed at considerably greater distances than found here for *D. excelsa* or *T. speciosissima*. Differences in dispersal distances between our study species and between sites for *T. speciosissima* in this study are probably related to the height of the infructescences and the height and density of the surrounding vegetation (Hammill *et al.* 1998). Infructescences of *D. excelsa* at Site 1 were on average 3.9 m (±0.15 s.e.) tall. For *T. speciosissima* at Site 1, infructescences were relatively shorter at 1.85 m (±0.08 s.e.), while *T. speciosissima* at Site 3 had the shortest infructescences at 0.95 m (±0.08 s.e.). The surrounding vegetation at Site 3 appeared to be denser than at Site 1 and this may have further contributed to the shorter dispersal distances. Dispersal along the ground by wind or water in heavy rains may also have contributed to these dispersal patterns (Harper 1977; Hammill *et al.* 1998). For serotinous species that release seeds shortly after fire, the understorey would be relatively open at the time of seed release, while for *D. excelsa* and *T. speciosissima*, considerable post-fire regrowth would have increased the density of the understorey. This may act to limit seed dispersal in *D. excelsa* and *T. speciosissima*.

The exclosure experiment indicated that seed losses to mammals and invertebrates occurred in the field (Fig. 2), as has been found for other large-seeded species in similar habitats (e.g. *Banksia* spp., *Isopogon anemonifolius* and *Petrophile pulchella*, Bradstock 1991; *Protea neriifolia*, Botha and Le Maitre 1992; *Grevillea* spp., Auld and Denham 1999, 2001; *Hakea sericea* and *P. pulchella*, R. J. Whelan and J. York, unpubl. data). Generally, other studies have found greatly reduced predation in burnt sites (Bradstock 1991; Botha and le Maitre 1992), while R. J. Whelan and J. York (unpubl. data) found that survival of seeds among exclusion treatments was extremely variable among fires and sites. The high level of losses observed in this study is more comparable to that found at unburnt sites (e.g. 100% losses, Bradstock 1991; 75–100% Botha and Le Maitre 1992; 82–95%, Auld and Denham 1999), and to that of recently burnt sites for *Grevillea caleyi* (Auld and Denham 2001). Most seed losses were presumably caused by mammals rather than invertebrates, on the basis of the patterns of seed removal and *in situ* damage. Losses were high for clumped seeds, but generally lower for seeds placed singly, implying that local seed density may affect the ability of seed predators to find

and subsequently eat seeds. Differences in seed losses between study species at Site 1 for low-density seeds may reflect a difference in palatability of seeds or degree of predator satiation, given the much greater seed production in *D. excelsa* (Table 1). At Site 3 for *T. speciosissima*, losses were high for seeds at high and low densities. While predator satiation may occur in the immediate post-fire period when species with canopy seed banks release seeds, it appears that such effects are less important by the time *D. excelsa* and *T. speciosissima* release seeds. Consequently, if recruitment is not limited by a lack of safe sites, then seed losses to predators are likely to have had an impact on the magnitude of seedling recruitment, particularly for *T. speciosissima* at Site 3.

Seedling recruitment was observed at all sites, but generally only a small percentage of the seeds released successfully established (≤4%, Table 3). The exception was at Site 1 for *T. speciosissima* where about 14% of all seeds released post-fire were established as seedlings. However, seedling densities by 1998 were comparable between *D. excelsa* and *T. speciosissima* at Site 1 (0.3 per m^2, Table 3), resulting from the much greater seed production of *D. excelsa* at this site (Table 1). In contrast, for *T. speciosissima* at Site 3, seedling densities were an order of magnitude lower (0.03 per m^2, Table 3), even though there was no significant difference in seed production to *T. speciosissima* at Site 1.

The highest levels of seedling recruitment in relation to the available seeds observed here (*T. speciosissima* at Site 1) were comparable to some estimates from local species with canopy seed banks, such as *Banksia serrata* (11–21% over 3 years) and *Isopogon anemonifolius* (6–18%) (Bradstock 1990), and from previous work on *T. speciosissima* (7–9%) (Bradstock 1995), while levels for *D. excelsa* and *T. speciosissima* (Site 3) were lower than those in these previous studies. In terms of individual plant fitness, the number of seedlings per flowering plant was more variable. For *T. speciosissima* at Site 1, the number of seedlings per flowering plant (22.7) was slightly greater than that found by Bradstock (1995) (range 1–9.7), while at Site 3 it was much lower (0.1). For *D. excelsa*, at three plots the number of seedlings per flowering plant (6–16.8) was comparable to that found by Bradstock (1995) for *T. speciosissima*, but at Site 1 plot 1 it was much greater (64.5).

The impact of low recruitment from one episode of post-fire reproduction depends on the ability of the recruits to survive to fire resistance (primary juvenile phase) and on the ability of the existing adults to survive future fires and produce additional seed crops (Bradstock 1990). Bradstock (1995) showed that less than 25% of *T. speciosissima* seedlings are resistant to fire at 8–10 years old. Further demographic studies should establish the length of the primary juvenile period of these species. Reduction in recruitment levels over the three sampled fruiting seasons in *T. speciosissima* in this study may be explained by variation in seed predation, environmental conditions or the availability of suitable sites for germination and establishment. This decline in the rate of establishment concurs with the accepted wisdom that conditions for recruitment decline with time since fire in these environments (Whelan 1995).

The distribution of seedlings in this study was not solely influenced by the shape of the seed shadow. There is a pattern for the seeds nearest to adults to have a relatively lower chance of successful establishment as seedlings. This pattern was most evident in *D. excelsa* (Fig. 1 *v.* 5*a*) and *T. speciosissima* at Site 1 (Fig. 1 *v.* 5*b*). Hutchings (1986) suggested that there are a number of factors that determine the final spatial distribution of seedlings, including the behaviour of seed predators and herbivores (Janzen 1970) and the distribution of safe sites, while Schupp and Fuentes (1995) warned that the correspondence of distributions or their lack cannot say much about what determines the ultimate distribution of adults. The distribution of safe sites is affected by competitive influences of adult plants (Sarukhan *et al.* 1985; Hutchings 1986; Crawley 1990; Le Maitre and Midgely 1992; Morgan 1995; Tozer and Bradstock 1997) and possibly allelopathy (Muller *et al.* 1968; Crawley 1986, 1990; Hutchings 1986). For *D. excelsa*, the leaves of adults protrude beyond the basal rosette and may reduce light and rainfall at the soil surface within the first 0.5 m of the rosette. For *T. speciosissima*, the lignotubers of adult plants extend beyond the aerial stems and result in reduced soil depth adjacent to the base of plants. Nevertheless, some seedlings were observed growing from the base of *D. excelsa* and from within the area of the lignotuber of *T. speciosissima*, although these seedlings may not persist. It is also apparent that our estimates of the seed shadow of plants did not detect some seeds that successfully germinated and became recruits well away from potential parents. These rarer, long-distance-dispersed seedlings may have greater significance for the species for colonisation of new habitat or recolonisation after adult mortality (Portnoy and Willson 1993). The distribution patterns observed here may change again after further mortality occurs, especially in relation to survival during future fires.

In this study, after just one fire, it is apparent that significant recruitment by *D. excelsa* and *T. speciosissima* occurred despite the long time between the fire and the release of seeds into the environment. The strategy of a single or few post-fire flowering events has the advantage that adults of these species may then redirect all their resources to surviving and flowering after the next fire.

Because of their spectacular inflorescences, some pyrogenic, flowering resprouters are subject to harvesting for the cut-flower market (e.g. *Blandfordia* spp., *D. excelsa*, *Telopea* spp. and *Xanthorrhoea* spp., NSW National Parks and Wildlife 2002). In NSW, south-eastern Australia, despite licensing requirements under the NSW National Parks and Wildlife Act 1987, little is known of the impact of

inflorescence-harvesting on the viability of populations of these species. Where harvesting of inflorescences of *D. excelsa* or *T. speciosissima* occurs in populations in the wild, such harvests may compound the impacts of seed predation and possibly limit seedling recruitment. Such populations may be at risk of decline if subject to excessive harvesting. Studies of other species subject to floral harvesting have indicated excessive harvesting leads to reduced seed production or seed-bank size (e.g. *Banksia hookeriana*, Witkowski *et al.* 1994: *Leucodendron* spp. and *Protea* spp., Mustart and Cowling 1992; Maze and Bond 1996), ultimately reducing the potential for the species to persist (Lamont *et al.* 2001). However, these are all fire-sensitive serotinous species, whose response may be quite different from pyrogenic flowering, resprouting species.

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